Chapter 4. Changes of Variables

Partitions of unity, the change of variables theorem, diffeomorphisms of $\mathbb{R}^n$, and its proof and applications

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Overview

The substitution rule of calculus, e.g. $\int_0^1 (2x^2+1)^{10}(4x)\,dx = \int_1^3 y^{10}\,dy$ with $y = 2x^2+1$, is generalized in two ways:

(1) to n-dimensional integrals, and (2) to the *extended* integral over open sets (no serious restriction, by Corollary 15.5). The result is the *change of variables theorem* (§17), proved in §19 using partitions of unity (§16) and the local factorization of diffeomorphisms into primitive ones (§18). §20 gives the geometric meaning of the determinant and the invariance of volume under isometries.

§16. Partitions of Unity

Instead of breaking the *set* A into compact rectifiable pieces C_N, we now break the *function* f into functions f_N each vanishing outside a compact set, and sum the integrals $\int_A f_N$. This approach recurs throughout the book.

Lemma 16.1

Let Q be a rectangle in $\mathbb{R}^n$. There is a C^∞ function $\phi : \mathbb{R}^n \to \mathbb{R}$ such that $\phi(\mathbf{x}) > 0$ for $\mathbf{x} \in \operatorname{Int} Q$ and $\phi(\mathbf{x}) = 0$ otherwise.

Proof. Let $f(x) = e^{-1/x}$ for $x > 0$ and $f(x) = 0$ otherwise. Then $f(x) > 0$ for $x > 0$, and f is of class C^∞ (see the remark below). Let $g(x) = f(x) \cdot f(1-x)$; g is C^∞, positive for $0 < x < 1$ and zero otherwise. For $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$ define

$$\phi(\mathbf{x}) = g\Big(\frac{x_1 - a_1}{b_1 - a_1}\Big) \cdot g\Big(\frac{x_2 - a_2}{b_2 - a_2}\Big) \cdots g\Big(\frac{x_n - a_n}{b_n - a_n}\Big).$$

■

Why f is C^∞ (supplied)

For $x > 0$, induction shows $f^{(k)}(x) = p_k(1/x)e^{-1/x}$ with p_k a polynomial. Since $t^m e^{-t} \to 0$ as $t \to \infty$ for every m, $f^{(k)}(x) \to 0$ as $x \to 0^+$; and $f^{(k)}(0) = 0$ follows inductively from $\lim_{x\to 0^+} f^{(k)}(x)/x = \lim_{t\to\infty} t\,p_k(t)e^{-t} = 0$.

Lemma 16.2

Let $\mathcal{A}$ be a collection of open sets in $\mathbb{R}^n$; let A be their union. Then there exists a countable collection $Q_1, Q_2, \ldots$ of rectangles contained in A such that:

(1) The sets $\operatorname{Int} Q_i$ cover A.

(2) Each Q_i is contained in an element of $\mathcal{A}$.

(3) Each point of A has a neighborhood that intersects only finitely many of the sets Q_i.

(1) and (2) are easy; (3), the *local finiteness condition*, is the point.

Proof.

Step 1. Let $D_1, D_2, \ldots$ be compact subsets of A with union A and $D_i \subset \operatorname{Int} D_{i+1}$ (Lemma 15.1); put $D_i = \emptyset$ for $i \le 0$. Let $B_i = D_i - \operatorname{Int} D_{i-1}$. B_i is bounded and closed (the intersection of the closed sets D_i and $\mathbb{R}^n - \operatorname{Int} D_{i-1}$), hence compact, and it is disjoint from the closed set D_{i-2} since $D_{i-2} \subset \operatorname{Int} D_{i-1}$. For each $\mathbf{x} \in B_i$ choose a closed cube $C_\mathbf{x}$ centered at $\mathbf{x}$, contained in A, disjoint from D_{i-2}, and small enough to lie in an element of $\mathcal{A}$. The interiors of the $C_\mathbf{x}$ cover B_i; choose finitely many whose interiors cover B_i, and call this finite collection $\mathcal{C}_i$.

Step 2. Let $\mathcal{C} = \mathcal{C}_1 \cup \mathcal{C}_2 \cup \cdots$, a countable collection of cubes, each contained in an element of $\mathcal{A}$. Given $\mathbf{x} \in A$, let i be the smallest integer with $\mathbf{x} \in \operatorname{Int} D_i$; then $\mathbf{x} \in B_i = D_i - \operatorname{Int} D_{i-1}$, so $\mathbf{x}$ lies interior to a cube of $\mathcal{C}_i$. Finally, every cube of $\mathcal{C}_{i+2}, \mathcal{C}_{i+3}, \ldots$ is disjoint from D_i by construction, so the open set $\operatorname{Int} D_i \ni \mathbf{x}$ meets only cubes of $\mathcal{C}_1, \ldots, \mathcal{C}_{i+1}$, finitely many. ■

Local finiteness holds at each point of A, but *not* at points of $\operatorname{Bd} A$: every neighborhood of such a point meets infinitely many cubes of $\mathcal{C}$.

Definition 16.1 (support)

If $\phi : \mathbb{R}^n \to \mathbb{R}$, the **support** of ϕ is the closure of $\{\mathbf{x} : \phi(\mathbf{x}) \neq 0\}$. Equivalently, if $\mathbf{x} \notin \operatorname{Support} \phi$, there is a neighborhood of $\mathbf{x}$ on which ϕ vanishes identically.

Theorem 16.3 (Existence of a partition of unity)

Let $\mathcal{A}$ be a collection of open sets in $\mathbb{R}^n$; let A be their union. There exists a sequence $\phi_1, \phi_2, \ldots$ of continuous functions $\phi_i : \mathbb{R}^n \to \mathbb{R}$ such that:

(1) $\phi_i(\mathbf{x}) \geq 0$ for all $\mathbf{x}$.

(2) The set $S_i = \operatorname{Support} \phi_i$ is contained in A.

(3) Each point of A has a neighborhood that intersects only finitely many of the sets S_i.

(4) $\sum_{i=1}^{\infty} \phi_i(\mathbf{x}) = 1$ for each $\mathbf{x} \in A$.

(5) The functions ϕ_i are of class C^∞.

(6) The sets S_i are compact.

(7) For each i, the set S_i is contained in an element of $\mathcal{A}$.

Definition 16.2 (partition of unity)

A collection $\{\phi_i\}$ satisfying (1)–(4) is a **partition of unity** on A. If it satisfies (5) it is **of class** C^∞; if (6), it has **compact supports**; if (7), it is **dominated by** the collection $\mathcal{A}$.

Proof. Let $Q_1, Q_2, \ldots$ be rectangles as in Lemma 16.2, and for each i let $\psi_i : \mathbb{R}^n \to \mathbb{R}$ be a C^∞ function positive on $\operatorname{Int} Q_i$ and zero elsewhere (Lemma 16.1). Then $\psi_i \geq 0$, $\operatorname{Support} \psi_i = Q_i$ is a compact subset of A contained in an element of $\mathcal{A}$, and each point of A has a neighborhood meeting only finitely many Q_i. So $\{\psi_i\}$ satisfies everything except (4). By (3), for $\mathbf{x} \in A$ only finitely many $\psi_i(\mathbf{x})$ are nonzero, so $\lambda(\mathbf{x}) = \sum_i \psi_i(\mathbf{x})$ converges trivially; near each point of A it is a finite sum of C^∞ functions, so λ is C^∞ on A. Also $\lambda(\mathbf{x}) > 0$ for $\mathbf{x} \in A$, since $\mathbf{x} \in \operatorname{Int} Q_i$ for some i. Define $\phi_i(\mathbf{x}) = \psi_i(\mathbf{x})/\lambda(\mathbf{x})$ (and 0 outside A; this is C^∞ on $\mathbb{R}^n$ since ψ_i vanishes outside the compact set $Q_i \subset A$). ■

By (1) and (4), the numbers $\phi_i(\mathbf{x})$ "partition unity" into non-negative pieces. Local finiteness implies that for any compact $C \subset A$, there is an open set about C on which all but finitely many ϕ_i vanish identically (cover C by finitely many such neighborhoods).

Example 16.1

Let $f(x) = (1 + \cos x)/2$ for $-\pi \leq x \leq \pi$ and $f(x) = 0$ otherwise; f is of class C^1. For $m \geq 0$ let $\phi_{2m+1}(x) = f(x - m\pi)$, and for $m \geq 1$ let $\phi_{2m}(x) = f(x + m\pi)$. Then $\{\phi_i\}$ is a partition of unity on $\mathbb{R}$: each support is a compact interval $[k\pi, (k+2)\pi]$, each point of $\mathbb{R}$ has a neighborhood meeting at most three supports, and $\sum \phi_i(x) = 1$ (since $f(x) + f(x - \pi) = 1$ on $[0, \pi]$, as $\cos(x - \pi) = -\cos x$).

Lemma 16.4

Let A be open in $\mathbb{R}^n$; let $f : A \to \mathbb{R}$ be continuous. If f vanishes outside the compact subset C of A, then the integrals $\int_A f$ and $\int_C f$ exist and are equal.

Proof. $\int_C f$ exists because C is bounded and f_C, which equals f on A and vanishes outside C, is continuous and bounded on $\mathbb{R}^n$. Let C_N be compact rectifiable sets with union A and $C_N \subset \operatorname{Int} C_{N+1}$. C is covered by finitely many $\operatorname{Int} C_N$, hence by one, $\operatorname{Int} C_M$. Since f vanishes outside C, $\int_C f = \int_{C_N} f$ for $N \geq M$. Applying this to $|f|$, $\lim \int_{C_N} |f|$ exists, so f is integrable over A (Theorem 15.2), and $\int_A f = \lim \int_{C_N} f = \int_C f$. ■

Theorem 16.5

Let A be open in $\mathbb{R}^n$; let $f : A \to \mathbb{R}$ be continuous. Let $\{\phi_i\}$ be a partition of unity on A having compact supports. The integral $\int_A f$ exists if and only if the series

$$\sum_{i=1}^{\infty} \Big[\int_A \phi_i |f| \Big]$$

converges; in this case,

$$\int_A f = \sum_{i=1}^{\infty} \Big[\int_A \phi_i f \Big].$$

Note that $\int_A \phi_i f$ exists and equals the ordinary integral $\int_{S_i} \phi_i f$, by Lemma 16.4.

Proof. First let $f \geq 0$.

Step 1. Suppose $\sum_i [\int_A \phi_i f]$ converges. Let D be a compact rectifiable subset of A. There is M such that ϕ_i vanishes identically on D for $i > M$, so $f(\mathbf{x}) = \sum_{i=1}^{M} \phi_i(\mathbf{x}) f(\mathbf{x})$ for $\mathbf{x} \in D$. Hence

$$\int_D f = \sum_{i=1}^{M} \Big[\int_D \phi_i f \Big] \leq \sum_{i=1}^{M} \Big[\int_{D \cup S_i} \phi_i f \Big] = \sum_{i=1}^{M} \Big[\int_A \phi_i f \Big] \leq \sum_{i=1}^{\infty} \Big[\int_A \phi_i f \Big],$$

by linearity, monotonicity and Lemma 16.4. So f is integrable over A and $\int_A f \leq \sum_i [\int_A \phi_i f]$.

Step 2. Suppose f is integrable over A. Given N, $D = S_1 \cup \cdots \cup S_N$ is compact, and for $i \leq N$, $\phi_i f$ vanishes outside D, so $\int_A \phi_i f = \int_D \phi_i f$. Thus

$$\sum_{i=1}^{N} \Big[\int_A \phi_i f \Big] = \int_D \Big[\sum_{i=1}^{N} \phi_i f \Big] \leq \int_D f \leq \int_A f,$$

using $\sum_{i \leq N} \phi_i \leq 1$. The partial sums are bounded, so the series converges with sum $\leq \int_A f$.

Step 3. For arbitrary continuous f: by Theorem 15.2, $\int_A f$ exists iff $\int_A |f|$ exists, iff (Steps 1, 2) $\sum [\int_A \phi_i |f|]$ converges. If $\int_A f$ exists,

$$\int_A f = \int_A f_+ - \int_A f_- = \sum_{i=1}^{\infty} \Big[\int_A \phi_i f_+ \Big] - \sum_{i=1}^{\infty} \Big[\int_A \phi_i f_- \Big] = \sum_{i=1}^{\infty} \Big[\int_A \phi_i f \Big],$$

since convergent series can be added term by term. ■

§17. The Change of Variables Theorem

Recall the convention $\int_b^a f = -\int_a^b f$.

Theorem 17.1 (Substitution rule)

Let $I = [a, b]$. Let $g : I \to \mathbb{R}$ be a function of class C^1, with $g'(x) \neq 0$ for $x \in (a, b)$. Then the set $g(I)$ is a closed interval J with end points $g(a)$ and $g(b)$. If $f : J \to \mathbb{R}$ is continuous, then

$$\int_{g(a)}^{g(b)} f = \int_a^b (f \circ g)g', \qquad \text{or equivalently,} \qquad \int_J f = \int_I (f \circ g)|g'|.$$

Proof. By continuity of g' and the intermediate-value theorem, $g' > 0$ on all of (a, b) or $g' < 0$ on all of (a, b); by the mean-value theorem g is strictly increasing or strictly decreasing, hence one-to-one, and $g(a) < g(b)$ or $g(a) > g(b)$ accordingly. Let $J = [c, d]$ be the interval with end points $g(a), g(b)$; by the intermediate-value theorem g carries I onto J. Let $F(y) = \int_c^y f$; then $F' = f$ by the fundamental theorem of calculus, and $h = F \circ g$ has $h'(x) = f(g(x))g'(x)$, which is continuous. Hence

$$\int_a^b (f \circ g)g' = h(b) - h(a) = F(g(b)) - F(g(a)) = \int_c^{g(b)} f - \int_c^{g(a)} f = \int_{g(a)}^{g(b)} f. \qquad (*)$$

If $g' > 0$, then $J = [g(a), g(b)]$ and $|g'| = g'$, so $(*)$ reads

$$\int_I (f \circ g)|g'| = \int_J f. \qquad (**)$$

If $g' < 0$, then $J = [g(b), g(a)]$ and $|g'| = -g'$, and $(*)$ again becomes $(**)$. ∎

Example 17.1

For $\int_{x=0}^{x=1} (2x^2 + 1)^{10}(4x)$, set $f(y) = y^{10}$, $g(x) = 2x^2 + 1$; $g'(x) = 4x > 0$ for $0 < x < 1$, so $\int_0^1 (2x^2 + 1)^{10}(4x) = \int_0^1 f(g(x))g'(x) = \int_1^3 y^{10}$.

Example 17.2

For $\int_{-1}^1 (1 - y^2)^{-1/2}$, calculus sets $y = g(x) = \sin x$, $-\pi/2 \leq x \leq \pi/2$; $g'(x) = \cos x > 0$ on $(-\pi/2, \pi/2)$, and formally

$$\int_{-1}^1 \frac{1}{(1 - y^2)^{1/2}} = \int_{-\pi/2}^{\pi/2} \frac{\cos x}{(1 - \sin^2 x)^{1/2}} = \int_{-\pi/2}^{\pi/2} 1 = \pi.$$

But the substitution rule does *not* apply: $f(y) = (1 - y^2)^{-1/2}$ is not continuous (not even bounded) on $[-1, 1]$; the integral is improper. This is one reason to prove the theorem for the extended integral.

Definition 17.1 (change of variables, diffeomorphism)

Let A be open in $\mathbb{R}^n$. A one-to-one function $g : A \to \mathbb{R}^n$ of class C^r with $\det Dg(\mathbf{x}) \neq 0$ for $\mathbf{x} \in A$ is a **change of variables** in $\mathbb{R}^n$. If A and B are open in $\mathbb{R}^n$ and $g : A \to B$ is a one-to-one map of A onto B such that g and g^{-1} are of class C^r, then g is a **diffeomorphism** (of class C^r).

These notions coincide. If g is a diffeomorphism, the chain rule shows Dg is non-singular, so g is a change of variables. Conversely, if $g : A \to \mathbb{R}^n$ is a change of variables, Theorem 8.2 shows $B = g(A)$ is open and $g^{-1} : B \to A$ is of class C^r.

Theorem 17.2 (Change of variables theorem)

Let $g : A \to B$ be a diffeomorphism of open sets in $\mathbb{R}^n$. Let $f : B \to \mathbb{R}$ be a continuous function. Then f is integrable over B if and only if the function $(f \circ g)|\det Dg|$ is integrable over A; in this case,

$$\int_B f = \int_A (f \circ g)|\det Dg|.$$

For $n = 1$, Dg is the 1 by 1 matrix $[g']$, so this contains the substitution rule; since the integrals are extended, it also justifies Example 17.2. The proof is in §19.

Example 17.3 (polar coordinates)

Let $B = \{(x, y) : x > 0,\ y > 0,\ x^2 + y^2 < a^2\}$. The **polar coordinate transformation** $g : \mathbb{R}^2 \to \mathbb{R}^2$, $g(r, \theta) = (r\cos\theta, r\sin\theta)$, has $\det Dg(r, \theta) = r$ and carries the open rectangle $A = \{(r, \theta) : 0 < r < a,\ 0 < \theta < \pi/2\}$ one-to-one onto B. Since $\det Dg = r > 0$ on A, $g : A \to B$ is a diffeomorphism, and

$$\int_B x^2y^2 = \int_A (r\cos\theta)^2(r\sin\theta)^2 r;$$

the latter exists as an ordinary integral and is evaluated by Fubini's theorem.

Example 17.4

To integrate x^2y^2 over $W = \{(x, y) : x^2 + y^2 < a^2\}$, note that g is not a diffeomorphism of an open set onto W. It is a diffeomorphism of $U = (0, a) \times (0, 2\pi)$ onto $V = \{(x, y) : x^2 + y^2 < a^2$, and $x < 0$ if $y = 0\}$, which is W with the non-negative x-axis deleted. Since that axis has measure zero, $\int_W x^2y^2 = \int_V x^2y^2$, which is expressed as an integral over U by polar coordinates.

Example 17.5 (spherical coordinates)

Let $B = \{(x, y, z) : x > 0,\ y > 0,\ x^2 + y^2 + z^2 < a^2\}$. The **spherical coordinate transformation** is

$$g(\rho, \phi, \theta) = (\rho\sin\phi\cos\theta,\ \rho\sin\phi\sin\theta,\ \rho\cos\phi), \qquad \det Dg = \rho^2\sin\phi,$$

positive for $0 < \phi < \pi$, $\rho \neq 0$. g carries $A = \{(\rho, \phi, \theta) : 0 < \rho < a,\ 0 < \phi < \pi,\ 0 < \theta < \pi/2\}$ one-to-one onto B, so

$$\int_B x^2 z = \int_A (\rho\sin\phi\cos\theta)^2(\rho\cos\phi)\rho^2\sin\phi,$$

evaluated by Fubini's theorem.

§18. Diffeomorphisms in $\mathbb{R}^n$

Two basic facts: the image of a compact rectifiable set under a diffeomorphism is compact and rectifiable; and any diffeomorphism factors locally into "primitive" diffeomorphisms.

Lemma 18.1

Let A be open in $\mathbb{R}^n$; let $g : A \to \mathbb{R}^n$ be a function of class C^1. If the subset E of A has measure zero in $\mathbb{R}^n$, then the set $g(E)$ also has measure zero in $\mathbb{R}^n$.

Proof.

Step 1. Let $\epsilon, \delta > 0$. If S has measure zero, S can be covered by countably many closed cubes, each of width less than δ, of total volume less than ϵ. It suffices to show that a rectangle $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$ can be covered by finitely many cubes of width $< \delta$ with total volume $< 2\,\mathrm{v}(Q)$. Choose $\lambda > 0$ with $Q_\lambda = [a_1 - \lambda, b_1 + \lambda] \times \cdots \times [a_n - \lambda, b_n + \lambda]$ of volume $< 2\,\mathrm{v}(Q)$. Choose N with $1/N$ less than both δ and λ. Among the numbers m/N ($m \in \mathbb{Z}$), let c_i be the largest with $c_i \le a_i$ and d_i the smallest with $d_i \ge b_i$; then $[a_i, b_i] \subset [c_i, d_i] \subset [a_i - \lambda, b_i + \lambda]$. The rectangle $Q' = [c_1, d_1] \times \cdots \times [c_n, d_n]$ contains Q, has $\mathrm{v}(Q') < 2\,\mathrm{v}(Q)$, and is partitioned by the points m/N into cubes of width $1/N < \delta$, whose total volume is $\mathrm{v}(Q')$ (Theorem 10.4).

Step 2. Let C be a closed cube in A with $|Dg(\mathbf{x})| \le M$ on C. If C has width w, then $g(C)$ lies in a closed cube of width $(nM)w$. Indeed, let $\mathbf{a}$ be the center of C. For $\mathbf{x} \in C$ the mean-value theorem gives $\mathbf{c}_j$ on the segment from $\mathbf{a}$ to $\mathbf{x}$ with $g_j(\mathbf{x}) - g_j(\mathbf{a}) = Dg_j(\mathbf{c}_j) \cdot (\mathbf{x} - \mathbf{a})$, so

$$|g_j(\mathbf{x}) - g_j(\mathbf{a})| \le n|Dg_j(\mathbf{c}_j)| \cdot |\mathbf{x} - \mathbf{a}| \le nM(w/2).$$

So $g(\mathbf{x})$ lies in the cube $\{\mathbf{y} : |\mathbf{y} - g(\mathbf{a})| \le nM(w/2)\}$ of width $(nM)w$.

Step 3. Let $E \subset A$ have measure zero. Let C_i be compact sets with union A and $C_i \subset \operatorname{Int} C_{i+1}$; with $E_k = C_k \cap E$ it suffices to show each $g(E_k)$ has measure zero. Given $\epsilon > 0$, choose $\delta > 0$ so that the δ-neighborhood of C_k (sup metric) lies in $\operatorname{Int} C_{k+1}$ (Theorem 4.6), and M with $|Dg(\mathbf{x})| \le M$ for $\mathbf{x} \in C_{k+1}$. By Step 1 cover E_k by countably many cubes of width $< \delta$ with total volume less than $\epsilon' = \epsilon/(nM)^n$; let $D_1, D_2, \ldots$ be those that actually meet E_k. Each D_i has width $< \delta$, so it lies in C_{k+1}, where $|Dg| \le M$; by Step 2, $g(D_i)$ lies in a cube D_i' of width nM(width D_i), so $\mathrm{v}(D_i') = (nM)^n\,\mathrm{v}(D_i)$. The cubes D_i' cover $g(E_k)$ with total volume less than $(nM)^n\epsilon' = \epsilon$. ∎

Example 18.1

Differentiability is needed: if g is merely continuous, the image of a set of measure zero need not have measure zero. There is a continuous map $f : [0,1] \to [0,1]^2$ whose image is the entire square (the *Peano space-filling curve*).

Theorem 18.2

Let $g : A \to B$ be a diffeomorphism of class C^r, where A and B are open sets in $\mathbb{R}^n$. Let D be a compact subset of A, and let $E = g(D)$.

(a) We have $g(\operatorname{Int} D) = \operatorname{Int} E$ and $g(\operatorname{Bd} D) = \operatorname{Bd} E$.

(b) If D is rectifiable, so is E.

Proof. (a) Since g^{-1} is continuous, g maps open sets of A to open sets of B (hence of $\mathbb{R}^n$). So $g(\operatorname{Int} D)$ is open and contained in $g(D) = E$:

$$g(\operatorname{Int} D) \subset \operatorname{Int} E. \tag{1}$$

Similarly g carries the open set $(\operatorname{Ext} D) \cap A$ onto an open set, disjoint from $g(D) = E$ since g is one-to-one:

$$g((\operatorname{Ext} D) \cap A) \subset \operatorname{Ext} E. \tag{2}$$

Hence

$$g(\operatorname{Bd} D) \supset \operatorname{Bd} E. \tag{3}$$

For let $\mathbf{y} \in \operatorname{Bd} E$. E is compact (continuous image of compact D), hence closed, so $\mathbf{y} \in E \subset B$; let $\mathbf{x} \in A$ with $g(\mathbf{x}) = \mathbf{y}$. By (1) $\mathbf{x} \notin \operatorname{Int} D$, and by (2) $\mathbf{x} \notin \operatorname{Ext} D$; so $\mathbf{x} \in \operatorname{Bd} D$. By symmetry the same holds for $g^{-1} : B \to A$: (1′) $g^{-1}(\operatorname{Int} E) \subset \operatorname{Int} D$, (3′) $g^{-1}(\operatorname{Bd} E) \supset \operatorname{Bd} D$. Combining (1),(1′) gives $g(\operatorname{Int} D) = \operatorname{Int} E$, and (3),(3′) give $g(\operatorname{Bd} D) = \operatorname{Bd} E$.

(b) If D is rectifiable, $\operatorname{Bd} D$ has measure zero; by Lemma 18.1 so does $g(\operatorname{Bd} D) = \operatorname{Bd} E$, and E is rectifiable. ■

Definition 18.1 (primitive diffeomorphism)

Let $h : A \to B$ be a diffeomorphism of open sets in $\mathbb{R}^n$ ($n \geq 2$), $h(\mathbf{x}) = (h_1(\mathbf{x}), \ldots, h_n(\mathbf{x}))$. h **preserves the ith coordinate** if $h_i(\mathbf{x}) = x_i$ for all $\mathbf{x} \in A$. If h preserves the ith coordinate for some i, h is a **primitive diffeomorphism**.

Theorem 18.3

Let $g : A \to B$ be a diffeomorphism of open sets in $\mathbb{R}^n$, where $n \geq 2$. Given $\mathbf{a} \in A$, there is a neighborhood U_0 of $\mathbf{a}$ contained in A, and a sequence of diffeomorphisms of open sets in $\mathbb{R}^n$,

$$U_0 \xrightarrow{h_1} U_1 \xrightarrow{h_2} U_2 \to \cdots \xrightarrow{h_k} U_k,$$

such that the composite $h_k \circ \cdots \circ h_2 \circ h_1$ equals $g|U_0$, and such that each h_i is a primitive diffeomorphism.

Proof.

Step 1. Linear case: $T(\mathbf{x}) = C \cdot \mathbf{x}$ with C non-singular factors into primitive non-singular linear maps. C is a product of elementary matrices (Theorem 2.4), whose transformations either (1) switch two coordinates, (2) replace the ith coordinate by itself plus a multiple of another, or (3) multiply the ith coordinate by a nonzero scalar. Types (2), (3) leave all but the ith coordinate fixed, so are primitive (for $n \geq 2$). Type (1) is a composite of types (2) and (3): starting from rows i, j equal to $\mathbf{a}, \mathbf{b}$, replace row i by row i − row j (giving $\mathbf{a}-\mathbf{b}, \mathbf{b}$), then row j by row j + row i (giving $\mathbf{a}-\mathbf{b}, \mathbf{a}$), then row i by row i − row j (giving $-\mathbf{b}, \mathbf{a}$), then multiply row i by -1 (giving $\mathbf{b}, \mathbf{a}$).

Step 2. A translation $t(\mathbf{x}) = \mathbf{x} + \mathbf{c}$ is the composite of $t_1(\mathbf{x}) = \mathbf{x} + (0, c_2, \ldots, c_n)$ and $t_2(\mathbf{x}) = \mathbf{x} + (c_1, 0, \ldots, 0)$, both primitive.

Step 3. Special case $\mathbf{a} = \mathbf{0}$, $g(\mathbf{0}) = \mathbf{0}$, $Dg(\mathbf{0}) = I_n$: g factors locally into two primitive diffeomorphisms. Define $h : A \to \mathbb{R}^n$ by $h(\mathbf{x}) = (g_1(\mathbf{x}), \ldots, g_{n-1}(\mathbf{x}), x_n)$. Then $h(\mathbf{0}) = \mathbf{0}$ and

$$Dh(\mathbf{x}) = \begin{bmatrix} \partial(g_1, \ldots, g_{n-1})/\partial \mathbf{x} \\ 0 \ \cdots \ 0 \ \ 1 \end{bmatrix};$$

since the first $n-1$ rows are those of Dg, $Dh(\mathbf{0}) = I_n$. By the inverse function theorem h is a diffeomorphism of a neighborhood V_0 of $\mathbf{0}$ with an open set V_1. Define $k : V_1 \to \mathbb{R}^n$ by $k(\mathbf{y}) = (y_1, \ldots, y_{n-1}, g_n(h^{-1}(\mathbf{y})))$. Then $k(\mathbf{0}) = \mathbf{0}$, and

$$Dk(\mathbf{y}) = \begin{bmatrix} I_{n-1} & 0 \\ D(g_n \circ h^{-1})(\mathbf{y}) \end{bmatrix}, \qquad D(g_n \circ h^{-1})(\mathbf{0}) = Dg_n(\mathbf{0}) \cdot [Dh(\mathbf{0})]^{-1} = [0 \ \cdots \ 0 \ 1],$$

so $Dk(\mathbf{0}) = I_n$ and k is a diffeomorphism of a neighborhood W_1 of $\mathbf{0}$ with an open set W_2. Let $W_0 = h^{-1}(W_1)$. The diffeomorphisms $W_0 \xrightarrow{h} W_1 \xrightarrow{k} W_2$ are primitive, and $k \circ h = g|W_0$: for $\mathbf{x} \in W_0$ and $\mathbf{y} = h(\mathbf{x}) = (g_1(\mathbf{x}), \ldots, g_{n-1}(\mathbf{x}), x_n)$,

$$k(\mathbf{y}) = (y_1, \ldots, y_{n-1}, g_n(h^{-1}(\mathbf{y}))) = (g_1(\mathbf{x}), \ldots, g_{n-1}(\mathbf{x}), g_n(\mathbf{x})) = g(\mathbf{x}).$$

Step 4. General case. Given $g : A \to B$ and $\mathbf{a} \in A$, let $C = Dg(\mathbf{a})$ and define $t_1(\mathbf{x}) = \mathbf{x} + \mathbf{a}$, $t_2(\mathbf{x}) = \mathbf{x} - g(\mathbf{a})$, $T(\mathbf{x}) = C^{-1} \cdot \mathbf{x}$. Let $\tilde{g} = T \circ t_2 \circ g \circ t_1$, a diffeomorphism of the open set $t_1^{-1}(A)$

with $T(t_2(B))$, satisfying $\tilde{g}(\mathbf{0}) = \mathbf{0}$ and $D\tilde{g}(\mathbf{0}) = C^{-1} \cdot Dg(\mathbf{a}) \cdot I_n = I_n$. By Step 3 there is an open set $W_0 \ni \mathbf{0}$ in $t_1^{-1}(A)$ on which $\tilde{g}$ factors into two primitive diffeomorphisms. Let $W_2 = \tilde{g}(W_0)$, $A_0 = t_1(W_0)$, $B_0 = t_2^{-1}T^{-1}(W_2)$. Then g carries A_0 onto B_0, and $g|A_0$ is the composite

$$A_0 \xrightarrow{t_1^{-1}} W_0 \xrightarrow{\tilde{g}} W_2 \xrightarrow{T^{-1}} T^{-1}(W_2) \xrightarrow{t_2^{-1}} B_0.$$

By Steps 1 and 2, t_1^{-1}, t_2^{-1} and T^{-1} factor into primitive transformations. ■

§19. Proof of the Change of Variables Theorem

We first prove the "only if" part.

Lemma 19.1

Let $g : A \to B$ be a diffeomorphism of open sets in $\mathbb{R}^n$. Then for every continuous function $f : B \to \mathbb{R}$ that is integrable over B, the function $(f \circ g)|\det Dg|$ is integrable over A, and

$$\int_B f = \int_A (f \circ g)|\det Dg|.$$

Proof.

Step 1. If the lemma holds for diffeomorphisms $g : U \to V$ and $h : V \to W$, it holds for $h \circ g$. Let $f : W \to \mathbb{R}$ be continuous and integrable over W. Then

$$\int_W f = \int_V (f \circ h)|\det Dh| = \int_U (f \circ h \circ g)\,|(\det Dh) \circ g|\,|\det Dg|;$$

the second integral exists and equals the first because the lemma holds for h, the third because it holds for g. It remains to note that by the chain rule $D(h \circ g)(\mathbf{x}) = Dh(g(\mathbf{x})) \cdot Dg(\mathbf{x})$, so $\det D(h \circ g) = [(\det Dh) \circ g] \cdot [\det Dg]$.

Step 2. Suppose each $\mathbf{x} \in A$ has a neighborhood $U \subset A$ such that the lemma holds for $g : U \to V$ ($V = g(U)$) and all continuous $f : V \to \mathbb{R}$ whose supports are compact subsets of V. Then the lemma holds for g. This is where partitions of unity are used. Write A as a union of such open sets U_α, with $V_\alpha = g(U_\alpha)$; their union is B. Choose a partition of unity $\{\phi_i\}$ on B with compact supports, dominated by $\{V_\alpha\}$. Then $\{\phi_i \circ g\}$ is a partition of unity on A with compact supports: $\phi_i(g(\mathbf{x})) \geq 0$; with $T_i = \operatorname{Support} \phi_i$, $g^{-1}(T_i)$ is compact and $\phi_i \circ g$ vanishes outside it, so $S_i = \operatorname{Support}(\phi_i \circ g) \subset g^{-1}(T_i)$ is compact; if W is a neighborhood of $g(\mathbf{x})$ meeting only finitely many T_i, then $g^{-1}(W)$ is a neighborhood of $\mathbf{x}$ meeting S_i only for these i; and $\sum_i \phi_i(g(\mathbf{x})) = 1$.

Let $f : B \to \mathbb{R}$ be continuous and integrable over B. By Theorem 16.5, $\int_B f = \sum_i [\int_B \phi_i f]$. Given i, choose α with $T_i \subset V_\alpha$. Since $\phi_i f$ is continuous on B and vanishes outside the compact set T_i, Lemma 16.4 gives $\int_B \phi_i f = \int_{T_i} \phi_i f = \int_{V_\alpha} \phi_i f$. By hypothesis for $g : U_\alpha \to V_\alpha$,

$$\int_{V_\alpha} \phi_i f = \int_{U_\alpha} (\phi_i \circ g)(f \circ g)|\det Dg|,$$

and since this integrand vanishes outside the compact set S_i, Lemma 16.4 gives $\int_B \phi_i f = \int_A (\phi_i \circ g)(f \circ g)|\det Dg|$. Summing,

$$\int_B f = \sum_{i=1}^{\infty} \Big[\int_A (\phi_i \circ g)(f \circ g)|\det Dg| \Big]. \qquad (*)$$

Since $|f|$ is integrable over B, $(*)$ holds with $|f|$ in place of f; as $\{\phi_i \circ g\}$ is a partition of unity on A, Theorem 16.5 shows $(f \circ g)|\det Dg|$ is integrable over A, and then $(*)$ and Theorem 16.5 give $\int_B f = \int_A (f \circ g)|\det Dg|$.

Step 3. The lemma holds for $n = 1$. Let $g : A \to B$ be a diffeomorphism of open sets in $\mathbb{R}$. Given $x \in A$, let I be a closed interval in A whose interior contains x, and $J = g(I)$, an interval with $g(\operatorname{Int} I) = \operatorname{Int} J$ (Theorems 17.1, 18.2). By Step 2 it suffices to verify, for continuous $f : \operatorname{Int} J \to \mathbb{R}$ with compact support in $\operatorname{Int} J$,

$$\int_{\operatorname{Int} J} f = \int_{\operatorname{Int} I} (f \circ g)|g'|. \qquad (**)$$

Extending f by 0 on $\operatorname{Bd} J$, $(**)$ is equivalent (Theorem 15.4 and Lemma 16.4) to $\int_J f = \int_I (f \circ g)|g'|$ for ordinary integrals, which is Theorem 17.1.

Step 4. Let $n > 1$. To prove the lemma for an arbitrary diffeomorphism $g : A \to B$ in $\mathbb{R}^n$, it suffices to prove it for primitive diffeomorphisms $h : U \to V$ of open sets in $\mathbb{R}^n$. Suppose so. Given $\mathbf{x} \in A$, Theorem 18.3 gives a neighborhood U_0 of $\mathbf{x}$ and primitive diffeomorphisms $h_1, \dots, h_k$ whose composite is $g|U_0$. By Step 1 the lemma holds for $g|U_0$; as $\mathbf{x}$ is arbitrary, Step 2 gives it for g.

Step 5. If the lemma holds in dimension $n-1$, it holds in dimension n. By Step 4 it suffices to treat a primitive diffeomorphism $h : U \to V$; say h preserves the last coordinate. Let $\mathbf{p} \in U$, $\mathbf{q} = h(\mathbf{p})$. Choose a rectangle $Q \subset V$ whose interior contains $\mathbf{q}$ and let $S = h^{-1}(Q)$; by Theorem 18.2, h is a diffeomorphism of $\operatorname{Int} S$ onto $\operatorname{Int} Q$. By Step 2 it suffices to prove the lemma for $h : \operatorname{Int} S \to \operatorname{Int} Q$ and continuous $f : \operatorname{Int} Q \to \mathbb{R}$ whose support is a compact subset of $\operatorname{Int} Q$. Then $(f \circ h)|\det Dh|$ vanishes outside a compact subset of $\operatorname{Int} S$, so is integrable over $\operatorname{Int} S$ (Lemma 16.4); we must show

$$\int_{\operatorname{Int} Q} f = \int_{\operatorname{Int} S} (f \circ h)|\det Dh|.$$

These integrals exist as ordinary integrals, so by Theorem 15.4 this is an equation in ordinary integrals. Extend f to $\mathbb{R}^n$ by 0 outside $\operatorname{Int} Q$, and define $F : \mathbb{R}^n \to \mathbb{R}$ by $F = (f \circ h)|\det Dh|$ on $\operatorname{Int} S$ and 0 elsewhere; both are continuous, and we must show $\int_Q f = \int_S F$.

Write $Q = D \times I$ with D a rectangle in $\mathbb{R}^{n-1}$ and I a closed interval. S is compact, so its projection to $\mathbb{R}^{n-1} \times 0$ lies in a set $E \times 0$ with E a rectangle in $\mathbb{R}^{n-1}$; since h preserves the last coordinate, $S \subset E \times I$. As F vanishes outside S, we must show $\int_Q f = \int_{E \times I} F$, which by Fubini's theorem is equivalent to

$$\int_{t \in I} \int_{\mathbf{y} \in D} f(\mathbf{y}, t) = \int_{t \in I} \int_{\mathbf{x} \in E} F(\mathbf{x}, t).$$

We show the inner integrals are equal. The slices of U, V at height t have the form $U_t \times t$, $V_t \times t$ with U_t, V_t open in $\mathbb{R}^{n-1}$; the slice of S is $S_t \times t$ with S_t compact. As $F(\cdot, t)$ vanishes outside S_t, and $f(\cdot, t)$ outside a compact subset of V_t, the needed equality is equivalent (Lemma 16.4) to

$$\int_{\mathbf{y} \in V_t} f(\mathbf{y}, t) = \int_{\mathbf{x} \in U_t} F(\mathbf{x}, t),$$

an equation in $(n-1)$-dimensional integrals. Now $h(\mathbf{x}, t) = (k(\mathbf{x}, t), t)$ for a C^1 function $k : U \to \mathbb{R}^{n-1}$, and

$$Dh = \begin{bmatrix} \partial k/\partial \mathbf{x} & \partial k/\partial t \\ 0 \ \cdots \ 0 & 1 \end{bmatrix}, \qquad \det Dh = \det \partial k/\partial \mathbf{x}.$$

For fixed t, $\mathbf{x} \mapsto k(\mathbf{x}, t)$ is a C^1 one-to-one map of U_t onto V_t with $\det \partial k/\partial \mathbf{x} \neq 0$, hence a diffeomorphism of open sets in $\mathbb{R}^{n-1}$. By the induction hypothesis,

$$\int_{\mathbf{y} \in V_t} f(\mathbf{y}, t) = \int_{\mathbf{x} \in U_t} f(k(\mathbf{x}, t), t)|\det \partial k/\partial \mathbf{x}|,$$

and for $\mathbf{x} \in U_t$ the integrand on the right is $f(h(\mathbf{x}, t))|\det Dh| = F(\mathbf{x}, t)$. ■

Now the "if" part.

Lemma 19.2

Let $g : A \to B$ be a diffeomorphism of open sets in $\mathbb{R}^n$; let $f : B \to \mathbb{R}$ be continuous. If $(f \circ g)|\det Dg|$ is integrable over A, then f is integrable over B.

Proof. Apply Lemma 19.1 to the diffeomorphism $g^{-1} : B \to A$ and the function $F = (f \circ g)|\det Dg|$, continuous and integrable over A: the function $(F \circ g^{-1})|\det Dg^{-1}|$ is integrable over B. It equals f: if $g(\mathbf{x}) = \mathbf{y}$, then $D(g^{-1})(\mathbf{y}) = [Dg(\mathbf{x})]^{-1}$ (Theorem 7.4), so

$$(F \circ g^{-1})(\mathbf{y}) \cdot |\det D(g^{-1})(\mathbf{y})| = F(\mathbf{x}) \cdot |1/\det Dg(\mathbf{x})| = f(\mathbf{y}).$$

■

Lemmas 19.1 and 19.2 together prove Theorem 17.2.

Example 19.1

If both integrals in the change of variables theorem exist as ordinary integrals, the theorem says they are equal. But it can happen that only one exists as an ordinary integral. In Example 17.2, the diffeomorphism $g : (-\pi/2, \pi/2) \to (-1, 1)$, $g(x) = \sin x$, gives

$$\int_{(-1,1)} 1/(1-y^2)^{1/2} = \int_{(-\pi/2,\pi/2)} 1;$$

the right side exists as an ordinary integral, but the left does not.

§20. Applications of Change of Variables

20.1 The meaning of the determinant

Theorem 20.1

Let A be an n by n matrix. Let $h : \mathbb{R}^n \to \mathbb{R}^n$ be the linear transformation $h(\mathbf{x}) = A \cdot \mathbf{x}$. Let S be a rectifiable set in $\mathbb{R}^n$, and let $T = h(S)$. Then

$$\mathrm{v}(T) = |\det A| \cdot \mathrm{v}(S).$$

Proof. If A is non-singular, h is a diffeomorphism of $\mathbb{R}^n$ with itself carrying $\operatorname{Int} S$ onto $\operatorname{Int} T$, and T is rectifiable (Theorem 18.2; for non-compact S apply it to $\bar{S}$). By the change of variables theorem (and Theorem 14.2(e)),

$$\mathrm{v}(T) = \mathrm{v}(\operatorname{Int} T) = \int_{\operatorname{Int} T} 1 = \int_{\operatorname{Int} S} |\det Dh| = \int_{\operatorname{Int} S} |\det A| = |\det A| \cdot \mathrm{v}(S).$$

If A is singular, $\det A = 0$ and we show $\mathrm{v}(T) = 0$. T is bounded, and h carries $\mathbb{R}^n$ onto a linear subspace V of dimension $p < n$, which has measure zero in $\mathbb{R}^n$. Then $\bar{T}$ is closed, bounded and of measure zero; 1_T is continuous except on $\bar{T}$ and vanishes outside $\bar{T}$, so $\int_T 1$ exists and equals 0. ■

So $|\det A|$ is the factor by which $h(\mathbf{x}) = A \cdot \mathbf{x}$ multiplies volumes.

Definition 20.1 (parallelopiped)

Let $\mathbf{a}_1, \ldots, \mathbf{a}_k$ be independent vectors in $\mathbb{R}^n$. The **k-dimensional parallelopiped** $\mathcal{P} = \mathcal{P}(\mathbf{a}_1, \ldots, \mathbf{a}_k)$ is the set of all $\mathbf{x} = c_1\mathbf{a}_1 + \cdots + c_k\mathbf{a}_k$ with $0 \le c_i \le 1$. The vectors $\mathbf{a}_i$ are the **edges** of $\mathcal{P}$.

A 2-dimensional parallelopiped is a parallelogram; a 3-dimensional one is the usual parallelopiped. For $k = n$ we already have a notion of volume:

Theorem 20.2

Let $\mathbf{a}_1, \dots, \mathbf{a}_n$ be n independent vectors in $\mathbb{R}^n$. Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$ be the n by n matrix with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$. Then

$$\mathrm{v}(\mathcal{P}(\mathbf{a}_1, \dots, \mathbf{a}_n)) = |\det A|.$$

Proof. $h(\mathbf{x}) = A \cdot \mathbf{x}$ carries $\mathbf{e}_j$ to $\mathbf{a}_j$ and the unit cube $I^n = [0,1]^n$ onto $\mathcal{P}(\mathbf{a}_1, \dots, \mathbf{a}_n)$. By Theorem 20.1, $\mathrm{v}(\mathcal{P}) = |\det A| \cdot \mathrm{v}(I^n) = |\det A|$. ■

Example 20.1

In calculus, the volume of the parallelopiped with edges $\mathbf{a}, \mathbf{b}, \mathbf{c}$ in $\mathbb{R}^3$ is, up to sign, the triple scalar product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \det[\mathbf{a} \ \mathbf{b} \ \mathbf{c}]$, and its sign depends on whether $\mathbf{a}, \mathbf{b}, \mathbf{c}$ is "right-handed" or "left-handed". We generalize this second notion.

Definition 20.2 (frame, orientation)

Let V be an n-dimensional vector space. An n-tuple $(\mathbf{a}_1, \dots, \mathbf{a}_n)$ of independent vectors in V is an **n-frame** in V. In $\mathbb{R}^n$, a frame is **right-handed** if $\det[\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] > 0$ and **left-handed** otherwise. The collection of all right-handed frames in $\mathbb{R}^n$ is an **orientation** of $\mathbb{R}^n$, and so is the collection of all left-handed frames. In general, choose a linear isomorphism $T : \mathbb{R}^n \to V$; one orientation of V consists of all frames $(T(\mathbf{a}_1), \dots, T(\mathbf{a}_n))$ with $(\mathbf{a}_1, \dots, \mathbf{a}_n)$ right-handed in $\mathbb{R}^n$, the other of those with $(\mathbf{a}_1, \dots, \mathbf{a}_n)$ left-handed. Each is the **reverse** (or **opposite**) of the other.

This is independent of the choice of T (by Theorem 20.3 below applied to $T'^{-1} \circ T$). In an arbitrary n-dimensional vector space "right-handed" is not well-defined, but "orientation" is.

Example 20.2

In $\mathbb{R}^1$ a frame is a single nonzero number; it is right-handed iff positive. In $\mathbb{R}^2$, $(\mathbf{a}_1, \mathbf{a}_2)$ is right-handed iff one must rotate $\mathbf{a}_1$ counterclockwise through an angle less than π to point it in the direction of $\mathbf{a}_2$. In $\mathbb{R}^3$, $(\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3)$ is right-handed iff curling the fingers of one's right hand from $\mathbf{a}_1$ to $\mathbf{a}_2$ makes the thumb point in the direction of $\mathbf{a}_3$. One justification: if a frame $(\mathbf{a}_1(t), \mathbf{a}_2(t), \mathbf{a}_3(t))$ varies continuously for $0 \le t \le 1$ and is right-handed at $t = 0$, it stays right-handed, since $\det[\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3]$ cannot change sign (intermediate-value theorem); and $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ satisfies both conditions.

Theorem 20.3

Let C be a non-singular n by n matrix. Let $h : \mathbb{R}^n \to \mathbb{R}^n$ be the linear transformation $h(\mathbf{x}) = C \cdot \mathbf{x}$. Let $(\mathbf{a}_1, \dots, \mathbf{a}_n)$ be a frame in $\mathbb{R}^n$. If $\det C > 0$, the frames $(\mathbf{a}_1, \dots, \mathbf{a}_n)$ and $(h(\mathbf{a}_1), \dots, h(\mathbf{a}_n))$ belong to the same orientation of $\mathbb{R}^n$; if $\det C < 0$, they belong to opposite orientations of $\mathbb{R}^n$.

If $\det C > 0$, h is **orientation-preserving**; if $\det C < 0$, **orientation-reversing**.

Proof. Let $\mathbf{b}_i = h(\mathbf{a}_i)$. Then $C \cdot [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_n]$, so $(\det C) \det[\mathbf{a}_1 \cdots \mathbf{a}_n] = \det[\mathbf{b}_1 \cdots \mathbf{b}_n]$. ■

20.2 Invariance of volume under isometries

Definition 20.3 (orthogonal and orthonormal sets)

Vectors $\mathbf{a}_1, \dots, \mathbf{a}_k$ of $\mathbb{R}^n$ form an **orthogonal set** if $\langle \mathbf{a}_i, \mathbf{a}_j \rangle = 0$ for $i \neq j$, and an **orthonormal set** if moreover $\langle \mathbf{a}_i, \mathbf{a}_i \rangle = 1$ for all i. If they are orthogonal and nonzero, the vectors $\mathbf{a}_i / \|\mathbf{a}_i\|$ are orthonormal.

An orthogonal set of nonzero vectors is independent: taking the dot product of $d_1\mathbf{a}_1 + \cdots + d_k\mathbf{a}_k = \mathbf{0}$ with $\mathbf{a}_i$ gives $d_i \langle \mathbf{a}_i, \mathbf{a}_i \rangle = 0$, so $d_i = 0$. Hence an orthogonal set of n nonzero vectors in $\mathbb{R}^n$ is a basis; $\mathbf{e}_1, \dots, \mathbf{e}_n$ is one, but there are many others.

Definition 20.4 (orthogonal matrix)

An n by n matrix A is **orthogonal** if its columns form an orthonormal set; equivalently, $A^{\mathrm{tr}} \cdot A = I_n$.

If A is orthogonal, A^{tr} is a left inverse, hence (Theorem 2.5) a right inverse: A is orthogonal iff A is non-singular and $A^{\mathrm{tr}} = A^{-1}$. Also $\det A = \pm 1$, since $(\det A)^2 = (\det A^{\mathrm{tr}})(\det A) = \det(A^{\mathrm{tr}} \cdot A) = \det I_n = 1$. The orthogonal matrices form a *group*:

Theorem 20.4

Let A, B, C be orthogonal n by n matrices. Then:
(a) $A \cdot B$ is orthogonal.
(b) $A \cdot (B \cdot C) = (A \cdot B) \cdot C$.
(c) There is an orthogonal matrix I_n such that $A \cdot I_n = I_n \cdot A = A$ for all orthogonal A.
(d) Given A, there is an orthogonal matrix A^{-1} such that $A \cdot A^{-1} = A^{-1} \cdot A = I_n$.

Proof. (a) $(A \cdot B)^{\mathrm{tr}} \cdot (A \cdot B) = B^{\mathrm{tr}} \cdot A^{\mathrm{tr}} \cdot A \cdot B = B^{\mathrm{tr}} \cdot B = I_n$.

(b) is immediate and (c) holds since I_n is orthogonal.

(d) Since $A^{\mathrm{tr}} = A^{-1}$, $I_n = A \cdot A^{\mathrm{tr}} = (A^{\mathrm{tr}})^{\mathrm{tr}} \cdot A^{\mathrm{tr}} = (A^{-1})^{\mathrm{tr}} \cdot A^{-1}$, so A^{-1} is orthogonal. ■

Definition 20.5 (orthogonal transformation, isometry)

$h(\mathbf{x}) = A \cdot \mathbf{x}$ is an **orthogonal transformation** if A is an orthogonal matrix; equivalently, h carries the basis $\mathbf{e}_1, \dots, \mathbf{e}_n$ to an orthonormal basis. A map $h : \mathbb{R}^n \to \mathbb{R}^n$ is a (euclidean) **isometry** if $\|h(\mathbf{x}) - h(\mathbf{y})\| = \|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

Theorem 20.5

Let $h : \mathbb{R}^n \to \mathbb{R}^n$ be a map such that $h(\mathbf{0}) = \mathbf{0}$.
(a) The map h is an isometry if and only if it preserves dot products.
(b) The map h is an isometry if and only if it is an orthogonal transformation.

Proof. (a) We have

$$\|h(\mathbf{x}) - h(\mathbf{y})\|^2 = \langle h(\mathbf{x}), h(\mathbf{x}) \rangle - 2\langle h(\mathbf{x}), h(\mathbf{y}) \rangle + \langle h(\mathbf{y}), h(\mathbf{y}) \rangle, \tag{1}$$

$$\|\mathbf{x} - \mathbf{y}\|^2 = \langle \mathbf{x}, \mathbf{x} \rangle - 2\langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{y} \rangle. \tag{2}$$

If h preserves dot products, the right sides are equal, so h is an isometry. Conversely, if h is an isometry, then $\|h(\mathbf{x}) - h(\mathbf{0})\| = \|\mathbf{x} - \mathbf{0}\|$ gives $\|h(\mathbf{x})\| = \|\mathbf{x}\|$; so the first and last terms of (1) and (2) agree, as do the left sides, whence $\langle h(\mathbf{x}), h(\mathbf{y}) \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$.

(b) If $h(\mathbf{x}) = A \cdot \mathbf{x}$ with A orthogonal, then $\langle h(\mathbf{x}), h(\mathbf{y})\rangle = h(\mathbf{x})^{\mathrm{tr}} \cdot h(\mathbf{y}) = \mathbf{x}^{\mathrm{tr}} \cdot A^{\mathrm{tr}} \cdot A \cdot \mathbf{y} = \mathbf{x}^{\mathrm{tr}} \cdot \mathbf{y}$, so h preserves dot products and is an isometry by (a). Conversely, let h be an isometry with $h(\mathbf{0}) = \mathbf{0}$, let $\mathbf{a}_i = h(\mathbf{e}_i)$ and $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$. By (a) the $\mathbf{a}_i$ are orthonormal, so A is orthogonal. Since the $\mathbf{a}_i$ form a basis, $h(\mathbf{x}) = \sum_i \alpha_i(\mathbf{x})\mathbf{a}_i$ for real functions α_i; orthonormality gives $\langle h(\mathbf{x}), \mathbf{a}_j\rangle = \alpha_j(\mathbf{x})$, while $\langle h(\mathbf{x}), \mathbf{a}_j\rangle = \langle h(\mathbf{x}), h(\mathbf{e}_j)\rangle = \langle \mathbf{x}, \mathbf{e}_j\rangle = x_j$. So $\alpha_j(\mathbf{x}) = x_j$ and $h(\mathbf{x}) = \sum_i x_i \mathbf{a}_i = A \cdot \mathbf{x}$. ■

Theorem 20.6

Let $h : \mathbb{R}^n \to \mathbb{R}^n$. Then h is an isometry if and only if it equals an orthogonal transformation followed by a translation, that is, if and only if h has the form $h(\mathbf{x}) = A \cdot \mathbf{x} + \mathbf{p}$, where A is an orthogonal matrix.

Proof. Let $\mathbf{p} = h(\mathbf{0})$ and $k(\mathbf{x}) = h(\mathbf{x}) - \mathbf{p}$. Then $\|k(\mathbf{x}) - k(\mathbf{y})\| = \|h(\mathbf{x}) - h(\mathbf{y})\|$, so k is an isometry iff h is. Since $k(\mathbf{0}) = \mathbf{0}$, k is an isometry iff $k(\mathbf{x}) = A \cdot \mathbf{x}$ with A orthogonal (Theorem 20.5), i.e. iff $h(\mathbf{x}) = A \cdot \mathbf{x} + \mathbf{p}$. ■

Theorem 20.7

Let $h : \mathbb{R}^n \to \mathbb{R}^n$ be an isometry. If S is a rectifiable set in $\mathbb{R}^n$, then the set $T = h(S)$ is rectifiable, and $\mathrm{v}(T) = \mathrm{v}(S)$.

Proof. $h(\mathbf{x}) = A \cdot \mathbf{x} + \mathbf{p}$ with A orthogonal, so $Dh(\mathbf{x}) = A$ and $|\det A| = 1$; by the change of variables theorem (as in Theorem 20.1), $\mathrm{v}(T) = |\det A| \cdot \mathrm{v}(S) = \mathrm{v}(S)$. ■